

Butterflies and Ants: A Relationship Forged by Fire

By Julia Wine

The endangered Fender's blue butterfly is declining as its prairie habitat degrades due to fire exclusion. These prairies rely on fire to keep trees out and promote the flowering of native plants. But fire also plays a surprising role in the symbiotic relationship between the butterfly and ants. So, scientists are studying how fire affects this mutualism and how we can save the Fender's blue butterfly by burning prairies in the Willamette Valley.



Two Fender's blue butterflies resting on a Kinkaid's lupine

A mutualistic relationship is an interaction between two species that benefits them both. Often, one species provides the other with food, offers protection, or helps with reproduction or digestion. In the case of the Fender's blue butterfly, the ants provide the butterfly larvae with protection from predators and parasites, and the butterfly larvae "reward ants with nutrient rich liquid secretions" full of sugars.



A Fender's blue butterfly larva

Scientists in the Willamette Valley in Oregon have been studying how fire impacts the relationship between the Fender's blue butterfly and ants. Their research found that fire history and temperature affect how ants interact with the butterfly larvae and the survival of the larvae. Specifically, ants tended to the butterfly larvae for longer periods of time and larval survival was higher on sites that were burned. The scientists also used open-top chambers to manipulate the temperature at certain sites and they discovered that butterfly larvae recruited ants more quickly in warmer temperatures.



A prescribed fire on a prairie in the Willamette Valley

Previous research found that the Fender's blue butterfly laid more eggs after the prairie had been burned. This research strengthens the case that fire is beneficial for this butterfly species, not only by

increasing reproduction but also by enhancing its mutualism with ants. These findings support the expansion of controlled burns as a tool for prairie restoration in the Willamette Valley and suggest that fire can play a key role in saving the Fender's blue butterfly.

Source: Warchola, N., C. Bastianelli, C. B. Schultz, and E. E. Crone. 2015. Fire increases ant-tending and survival of the Fender's blue butterfly larvae. Journal of Insect Conservation 19:1063–1073.